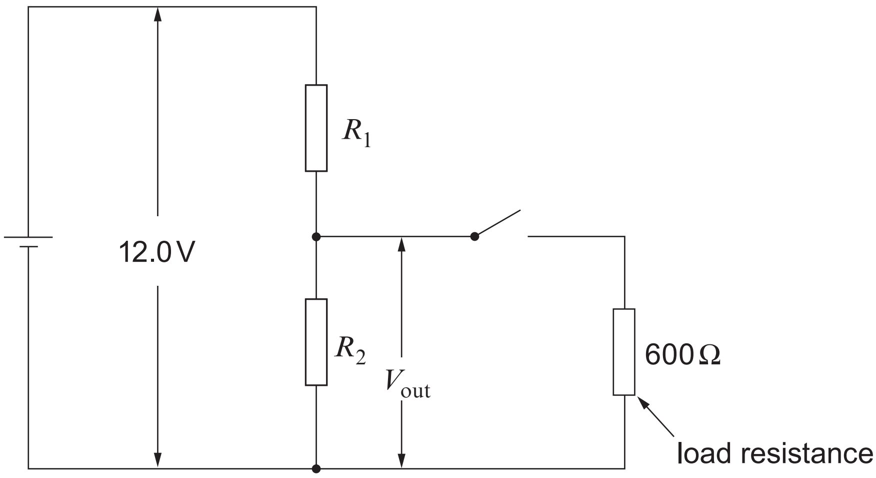
Examiner only

3. For the potential divider circuit shown a student is asked to select values for R_1 and R_2 so that the circuit meets certain requirements.



Requirements

A an output pd, V_{out}, of 3.0 V when unloaded (switch open),
B a power dissipation of less than 0.50 W in R_1 when unloaded (switch open),
C a **decrease** of less than 0.40 V in V_{out} when the switch is closed.

Showing your working clearly, determine whether or not the circuit meets the requirements when R_1 = 450 Ω and R_2 = 150 Ω.

(a) Requirement **A**: an output pd, V_{out}, of 3.0 V when unloaded (switch open). [2]

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(b) Requirement **B**: a power dissipation of less than 0.50 W in R_1 when unloaded (switch open). [2]

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(c) Requirement **C**: a **decrease** of less than 0.40 V in V_{out} when the switch is closed. [4]

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